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Type of early bilingualism and its effect on the acoustic realization of allophonic variants: Early sequential and simultaneous bilinguals

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Abstract

Aims and Objectives/Purpose/Research Questions: This study examines the acoustic realization of phrase-initial voiced stops (/b, d, g/) and intervocalic voiced approximants ([β, ð, ɣ]) in Spanish by three groups of Spanish–English bilinguals: simultaneous bilinguals, early sequential bilinguals, and late second language (L2) Spanish learners to investigate if the type of early bilingualism has an effect on the acoustic realization of language-specific phonological processes.

Design/Methodology/Approach: Early bilinguals were divided into two groups (simultaneous and sequential bilinguals), and together with late learners (L2 Spanish learners) they were recorded producing words eliciting phrase-initial and intervocalic voiced stops.

Data and Analysis: The acoustic analyses examined the difference between the intensity minimum during the consonant and the intensity maximum of the following tautosyllabic vowel (intensity difference). The dataset comprised a total of 3,569 acoustic measurements.

Findings/Conclusions: The group of early sequential bilinguals shows a significantly stronger tendency to produce a more lenited intervocalic segment than simultaneous bilinguals and L2 learners, who produce a more constricted realization. These data indicate that simultaneous bilinguals and L2 learners, who were exposed to English since birth, differ from early bilinguals who were exclusively exposed to Spanish for the first years of their life when applying the Spanish allophonic rule governing the distribution of voiced stops and spirants.

Originality: This is one of the first studies that investigates the effects of the type of early bilingualism on bilingual phonetic performance in adulthood, especially in terms of bilingual phonetic production. These novel data examine the acoustic realization of Spanish heritage speakers, a largely understudied bilingual population.

Significance/Implications: The findings suggest that the type of early bilingualism in language development, even at a very early age, still has consequences on the acoustic realization of allophonic variants in adulthood.

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Speech production, simultaneous bilingualism, sequential bilingualism, spirantization, heritage speakers, type of early bilingualism

Introduction

Previous studies investigating age of acquisition effects have consistently shown that cross-linguistic influence is minimized the earlier the language is acquired. These studies have demonstrated that early bilinguals generally outperform late bilinguals in terms of their phonetic abilities, as late learners typically face more difficulties in producing and perceiving native-like targets and are prone to exhibiting accent transfer from their L1 (DeKeyser, 2000; Flege, 1991, 1999, 2003, 2007; Lenneberg, 1967; Long, 1990; Oyama, 1976; Patkowski, 1990; Scovel, 1988; Yeni-Komshian, Flege, & Liu, 2000). This is not a surprising outcome based on language acquisition studies demonstrating that children become attuned to the sounds of their L1 at a very young age: over the course of the first year of life, monolingual and bilingual infants are considered to be generalized listeners, capable of discriminating between both native and non-native sound contrasts. However, during their first year of life, there is early phonological attunement, and these infants become specialized listeners, whose language discrimination is closely restricted to the sound system of their own native language or languages (Bosch & Sebastián-Gallés, 2003; Kuhl et al., 2008; Maye, Weiss, & Aslin, 2008; Werker & Tees, 1984).

For bilingual children exposed to two languages early in life, both languages will wax and wane over the years, as the relationship between both languages is dynamic. Therefore, the language(s) a person is exposed to during early childhood will affect how subsequent linguistic systems are learned and the cumulative linguistic experience of the bilingual individual will condition the various degrees of interlingual influence in the sound system of both languages at each stage of development during childhood and adulthood. In order to better understand the phonological/phonetic systems of early bilinguals, researchers necessarily have to consider many variables, such as language proficiency, language dominance, language use, and other biographical non-linguistic variables that are particular to the experiences of the bilingual community under investigation (Amengual, 2016a, 2016b, 2016c; Amengual & Chamorro, 2015; Baker & Trofimovich, 2005; Flege, Frieda, & Nozawa, 1997; Flege, Schirru, & MacKay, 2003; Guion, 2003; Mora & Nadeu, 2012; Simonet, 2014).

Heritage language phonetics and phonology

The present study investigates the phonetic production of two groups of *heritage speakers*, an understudied group of early bilinguals in the domain of phonetics/phonology. Benmamoun, Montrul, and Polinsky (2013) characterize heritage speakers as asymmetrical bilinguals who learned the heritage language as a L1 in childhood, but who, as adults, are dominant in a different language. In the context of the United States, the term heritage speaker has been used to refer to an individual who is raised in a home where a non-English language is spoken, who speaks or merely understands the heritage language, and who is to some degree bilingual in English and the heritage language (Valdés, 2000, 2005). The linguistic abilities of heritage speakers have been compared in recent years to both monolingual speakers and to L2 learners showing that heritage language acquisition typically results in transfer from the dominant language, a better ability with receptive than productive language, and linguistic gaps that resemble the patterns attested in L2 acquisition (Montrul, 2011; Montrul, Bhatt, & Bhatia, 2012; Polinsky, 2006; Rothman, 2007). Even though the phonetic abilities of heritage

speakers have recently received more attention, there are still considerably fewer studies on the production, perception, and processing of the heritage language, especially in comparison to research focusing on heritage language morphosyntactic knowledge or studies aimed at devising pedagogy tailored to meet heritage language learners' linguistic needs (Potowski, 2013; Rao & Ronquest, 2015). Heritage language researchers have urged investigators to experimentally test the assumption that heritage speakers maintain 'good phonology' in their minority language (Polinsky & Kagan, 2007), and these heritage speakers can provide new, valuable data for testing and advancing theoretical models of bilingual phonological acquisition.

The current study contributes to this effort by investigating the phonetic production abilities of Spanish heritage speakers and late-onset L1 English–L2 Spanish bilinguals in northern California (USA). Spanish heritage speakers, like other early bilinguals, are a heterogeneous group with marked differences in their language experience. Some heritage speakers have been exposed to the minority (heritage) language and the majority language early in life, either by growing up speaking both languages since birth (i.e., simultaneous bilinguals) or having been brought up in a monolingual setting in early childhood and becoming bilingual after starting school in the majority language at around ages 5 or 6 (i.e., early sequential bilinguals). In simultaneous bilingualism, the child acquires two languages at the same time in a process of bilingual L1 acquisition (BFLA; De Houwer, 1990, 2009; Meisel, 1989) and is considered to have two L1s, whereas early sequential (successive or consecutive) bilingualism refers to a process of early L2 acquisition (ESLA; De Houwer, 1990, 2009), in which the child has already partially acquired a L1 when learning a L2 early in childhood.

While research on the phonetics of heritage Spanish is beginning to receive more attention (e.g., Alvord and Rogers (2014), Ronquest (2012) for vowels; Amengual (2016c), Henriksen (2015) for rhotics; and Kim (2016), Rao (2016), Robles-Puente (2014), for suprasegmentals), we still do not have a clear understanding of how individuals produce their heritage language as a function of their type of early bilingualism. The question that remains unanswered is if acquiring both languages from birth, as in the context of simultaneous bilingualism or acquiring the L2 at a somewhat later age in the case of early sequential bilingualism has an effect on the acoustic realization of language-specific phonological processes in the heritage language.

Early bilingualism: A sensitive period for phonetic category formation?

Recent studies on Spanish–Catalan bilingual adults born and raised in Barcelona (Spain) have pointed to the effects of sequential learning on the phonetic behavior of early and highly proficient bilinguals. For instance, Sebastián-Gallés, Echeverría, and Bosch (2005) analyzed the perceptual abilities of two groups of early sequential Catalan–Spanish bilinguals in a lexical decision task. The Catalan–Spanish group had been primarily exposed to Catalan since birth and started learning Spanish at the beginning of mandatory schooling at age four. The Spanish–Catalan group had acquired Spanish since birth, and Catalan was learned at the onset of bilingual schooling. The results revealed that the Catalan–Spanish sequential bilinguals performed more accurately in the Catalan lexical decision task than the Spanish–Catalan sequential bilinguals. These findings suggest that the L1 (Catalan) enhanced the ability of the Catalan–Spanish sequential group to perform in Catalan whereas the L1 (Spanish) of the Spanish–Catalan sequential bilinguals affected their accuracy in the Catalan lexical decision task. When the results for early sequential bilinguals were compared to those of simultaneous bilinguals, the findings revealed that the Catalan–Spanish sequential bilinguals displayed a higher accuracy rate than the simultaneous bilingual group and the Spanish–Catalan sequential group. These results showed that language exposure during the earliest stages of language development affected their lexical representations influencing their perceptual abilities as adults.

Evidence of a sensitive period for phonetic category formation has even been found in individuals who do not know or use the language they were exposed to during early childhood. Research on international adoptees of Korean origin that were raised by Swedish families (Hyltenstam, Bylund, Abrahamsson, & Park, 2009) and on overhearers of Spanish who were not able to speak Spanish but who were passively exposed to the language as children (Au, Oh, Knightly, Jun & Romo, 2008; Knightly, Jun, Oh, & Au, 2003) has shown that individuals who, to the detriment of their L1, became dominant in their L2 can access residual L1 knowledge. In the case of Korean adoptees, Hyltenstam et al. (2009) found that the later the time of adoption, the better chances were for access to the L1 remnants in the process of re-exposure to the language. This could be explained by considering that these individuals who have no conscious recollection of their L1 had received more input in the language that contributed to forming language-specific phonological categories providing phonological advantages in the language, which were available during adulthood. Based on these studies, it is reasonable to hypothesize that the input that a bilingual individual receives during the early stages of development, as a result of either simultaneous bilingualism (BFLA) or early sequential bilingualism (ESLA), could shape language-specific phonemic categories and phonological processes that would be carried into adulthood.

The phonetic variable: Spirantization of Spanish /b, d, g/

This study examines a phonological difference between Spanish and English: articulatory reduction (lenition), known as spirantization, wherein the voiced stops /b, d, g/ in Spanish are articulated with full occlusion only phrase initially, after a homorganic nasal, and for /d/ only when there is a preceding /l/. In all other positions these voiced stops are commonly realized as the voiced approximants [β, ð, ɣ][1] (Harris, 1969; Mascaró, 1984). Table 1 provides the distribution of Spanish stop and approximant allophones of /b/, /d/, and /g/ as a function of the phonological contexts that favor either a stop [b, d, g] or approximant [β, ð, ɣ] realization.

Voiced stops /b, d, g/ systematically surface as [β, ð, ɣ] in intervocalic position. This phonological rule applies to /b, d, g/ appearing in intervocalic contexts both within a word (e.g., the /b/ in *caballo* [ka'βaʎo] 'horse') and across word boundaries (e.g., the /b/ in *la boca* [la 'βoka] 'the mouth'). Previous studies have found a great deal of inter-speaker and intra-speaker variability in the intervocalic articulatory weakening of /b, d, g/ in the speech of native speakers of Spanish due to phonetic, prosodic, and social factors (Carrasco, Hualde, & Simonet, 2012; Eddington, 2011; Hualde, Simonet, & Nadeu, 2011; Ortega-Llebaría, 2004; Rogers, 2016).

Table 1. Distribution of Spanish [b, d, g] and [β, ð, ɣ] as a function of phonological context.

Phonological context	Realization	Examples
Phrase-initially (after a pause)	Stop [b, d, g]	*boca* [boka] 'mouth', *dos* [dos] 'two', *gota* [gota] 'drop'
Following a nasal consonant	Stop [b, d, g]	*ambos* [ambos] 'both', *andar* [andar] 'walk', *mango* [maŋgo] 'mango'
Following a lateral consonant	Stop [d] and approximant [β, ɣ]	*caldo* [kaldo] 'soup', *alba* [alβa] 'dawn', *algo* [alɣo] 'something'
Following other consonants (not nasal or lateral)	Approximants [β, ð, ɣ]	*árbol* [arβol] 'tree', *desde* [dezðe] 'since/from', *largo* [larɣo] 'long'
Intervocalic	Approximants [β, ð, ɣ]	*saber* [saβer] 'know', *codo* [koðo] 'elbow', lago [laɣo] 'lake'

Previous studies have suggested that there are a number of phonetic factors that contribute to the variation in the acoustic realization of intervocalic approximants in Spanish. It should be noted that there is an expected longer and more constricted production at the onset of a stressed syllable compared with an unstressed syllable (Carrasco et al., 2012; Cole, Hualde, & Iskarous, 1999; Eddington, 2011; Ortega-Llebaría, 2004). A higher rate of [ð] deletion has also been observed in highly frequent words, and reduction of [ð] has also been reported to be higher when it appears in the past participle suffix (Bybee, 2002). In terms of surrounding vowels, Ortega-Llebaría (2004) observed more lenition for [ɣ] when surrounded by [a] than by [i] or [u], while no differences between surrounding vowels were found for [β]. In addition to phonetic factors, several studies have investigated the intervocalic production of /b, d, g/ from a sociolinguistic perspective. For instance, intervocalic /d/ is deleted at higher rates by speakers of Dominican Spanish with a lower socioeconomic status, and this is particularly the case of younger speakers (Alba, 2000). In a study on the intervocalic production of /b, d, g/ in the Spanish of Yucatán, Michnowicz (2011) observed that women, older speakers, and those with monolingual Maya-speaking parents favored the production of intervocalic stops over approximants.

Spirantization has been shown to be a difficult process for native English speakers acquiring Spanish as a L2 (Face & Menke, 2009; González-Bueno, 1995; Zampini, 1994) since spirants are not allophonic variants of stops in English. Even though English speakers may spirantize both voiced and voiceless stops in fast, casual speech (Brown, 1990; Gimson, 1989), spirants are much less frequent and primarily an effect of speech style. In contrast to English, this lenition rule is a fully conventionalized process systematically found in all speech styles in Spanish, including careful reading style (Hualde et al., 2011). Recent L2 research has investigated the ability of English speakers to produce the voiced approximants focusing on the acoustic realization of late L2 Spanish learners with different levels of instruction (Alvord & Christiansen, 2012; Cabrelli Amaro, 2017; Face & Menke, 2009; Rogers & Alvord, 2014; Shea & Curtin, 2010; Zampini, 1994) and Spanish heritage speakers (Rao, 2014, 2015). However, we still do not know if there are differences between early sequential and simultaneous bilinguals with respect to their acoustic realization of language-specific phonological processes in the heritage language, such as the distribution of voiced stops and spirants in Spanish.

The present study

Focusing on the acoustic realization of language-specific phonological alternations (i.e., Spanish spirantization), the present study relies on intensity measurements to operationalize the degree of lenition (spirantization) of /b, d, g/ in phrase-initial and intervocalic position, as carried out in previous studies (Carrasco et al., 2012; Cole et al., 1999; Eddington 2011; Ortega-Llebaría, 2004), in order to answer the following question: do simultaneous and early sequential bilinguals differ in their production of allophonic variants in their heritage language (i.e., Spanish) as a result of differences in the type of early bilingualism? If the type of early bilingualism affects the acoustic realization of the stop/approximant allophonic alternation in Spanish, we would expect to find differences in the rates of lenited/spirantized consonants in intervocalic contexts between early simultaneous, early sequential bilinguals, and late L2 learners.

Method

Participants

Thirty female undergraduate students enrolled in a public university in northern California were recruited to participate in the study. The sample consisted of 20 Spanish heritage speakers and 10

Table 2. Age, age of exposure, accent self-ratings, and typical daily use of both languages for each group.

	Simultaneous bilinguals	Sequential bilinguals	Second language Spanish learners
	M (*SD*)	M (*SD*)	M (*SD*)
Age	20.5 (0.5)	19.8 (1.1)	20.8 (0.7)
Age of exposure	E = 0.4 (0.3) S = 0.3 (0.2)	E = 4.8 (1.8) S = 0.2 (0.3)	E = 0.0 (0.0) S = 8.6 (4.8)
Self-reported accent (1 = non-native, 10 = native)	E = 8.8 (1.1) S = 7.2 (0.9)	E = 8.2 (1.4) S = 7.9 (1.3)	E = 9.6 (0.6) S = 6.2 (1.5)
Daily use (1 = only English, 9 = only Spanish)	5.1 (1.7)	5.3 (2.1)	2.8 (1.4)

M: Mean; S: Standard; D: Deviation.

L2 Spanish learners. Based on their answers to the question "at what age did you start learning English/Spanish?" together with follow-up questions on the language(s) spoken with their parents and other family members, their accent self-ratings, typical daily use, and their age of exposure to each language, participants were divided into three groups: early sequential bilinguals; early simultaneous bilinguals; and late L2 Spanish learners. Table 2 provides the age, age of exposure, accent self-ratings, and typical daily use of Spanish and English for each speaker group.

Both groups of Spanish heritage speakers (early sequential bilinguals and simultaneous bilinguals) had been raised and educated in a bilingual environment in the United States, having extensive exposure to both Spanish and English on a daily basis. Following the categorization of Silva-Corvalán (1994), these Spanish heritage speakers would be classified as members of the third immigrant generation (G3), in that they were born in the US, with at least one parent being a generation 2 (G2) speaker. The sequential bilingual group had been raised speaking Spanish at home with Spanish-speaking parents, had learned English in pre-school or kindergarten (by age 4–5), and had been raised and educated in the US. These sequential bilinguals had both parents from generation 2 (G3:2, Escobar & Potowski, 2015). The simultaneous bilinguals had been raised speaking both Spanish and English at home, reported the "one parent one language approach" at home, and were also educated in English in the US. Seven of these bilinguals grew up speaking Spanish with their mother and extended family and the remaining three simultaneous bilinguals spoke Spanish with their father and their extended family. The L2 Spanish learners had been raised in a monolingual English household in the United States, spoke English as their native language, and learned Spanish at school. All 30 participants were Spanish Studies or Language Studies majors with a concentration in Spanish, they had completed their entire education in the US, and they were enrolled in upper-division Spanish courses at the time of testing. These bilinguals would be considered to have at least an upper-intermediate level of university-level language study in Spanish.

All participants completed the Bilingual Language Profile (BLP) questionnaire (Birdsong, Gertken, & Amengual, 2012). The BLP is an instrument for assessing language dominance through self-reports and it produces a continuous dominance score and a general bilingual profile taking into account multiple dimensions: age of acquisition of the L1 and L2; frequency and contexts of use; competence in different skills; and attitudes towards each language (Gertken, Amengual, & Birdsong, 2014). All of these factors are organized in four modules, which receive equal weighting (language history, language use, language proficiency, and language attitudes).

The responses to the questionnaire generate a global score for each language (Spanish and English), a language particular score for each module, and a global score of dominance. The point system was converted to a scale score with the English score subtracted from the Spanish score. Based on their responses to the BLP, all participants were English-dominant, but the L2 Spanish learners were more English-dominant than the sequential bilinguals (*diff.* = 69.4, $t(9) = 14.4$, $p < 0.001$) and simultaneous bilinguals (*diff.* = 66.2, $t(9) = 12.2$, $p < 0.001$). However, there were no significant differences in the language dominance scores of simultaneous and sequential bilinguals (*diff.* =-3.1, $t(9) = -0.61$, not significant (n.s)). Figure 1 provides the distribution of the sequential bilinguals, simultaneous bilinguals, and L2 Spanish speakers as a function of their BLP dominance scores. Note that higher values indicate a more unbalanced English-dominance.

Materials and recording procedure

The production of the target voiced stops in phrase-initial initial and intervocalic position was elicited in a reading-aloud task. The materials consisted of two blocks of 60 Spanish sentences. Each sentence contained a target word within the carrier phrase, *Yo puedo decir* TARGETWORD "I can say TARGETWORD". The Spanish utterances in each repetition block elicited the pronunciation of 30 word-initial voiced stops and 30 word-medial intervocalic voiced stops (10 bilabial, 10 dental and 10 velar stops) in stressed position. The word-initial voiced stops occurred after a pause and were followed by a mid-vowel (*boca* 'mouth'), and the word-medial voiced stops were also followed by a mid-vowel (*abono* 'fertilizer', *adobo* 'marinade', *agosto* 'August'). It is important to note that the participants made a pause at the word boundary, which enabled this phrase-initial position to elicit the voiced stop realization. In addition to the experimental items, there were 70 fillers/distractors in each block.

The experiment was conducted with individual subjects seated comfortably in a sound-attenuated booth and facing a computer display. Participants were asked to read the sentences clearly and at a natural pace. The speech samples were recorded using a head-mounted microphone (Shure SM10A) and an audio interface (MOTU Ultra Like mk3), digitized (44kHz, 16-bit quantization), and computer-edited for subsequent acoustic analysis. Each participant produced 120 target productions (2 repetitions in randomized order of each of the 60 experimental items) for a total of 3,600 measurements. 31 tokens were excluded due to recording errors, or mispronunciations. The resulting dataset comprised a total of 3,569 tokens.

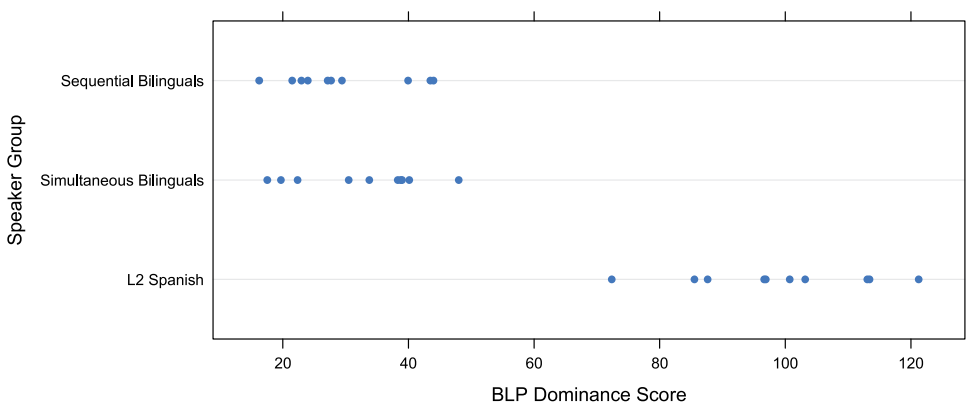

Figure 1. Language dominance as a function of group according to the Bilingual Language Profile.

Acoustic analysis

Two gradient measures of intensity from the target phrase-initial and intervocalic voiced stops were taken using Praat (Boersma & Weenink, 2016): the first was the lowest point of the intensity of the voiced stop/approximant; and the second was the intensity peak of the next vowel following the target voiced stop/approximant. For each target item, the valley of the consonant and the peak of the following vowel on the intensity curve were annotated (see Figure 2). All points were manually marked, and a Praat script was run to extract all points and subtract all the valleys from their corresponding peaks. Measuring the lowest point of intensity of the target stop or approximant, rather than the average intensity across the entire approximant was done to avoid the complications that arise when one tries to objectively determine the onset and offset of a highly lenited approximant (Eddington, 2011).

The degree of spirantization was determined by the degree of oral occlusion between the consonant and the following vowel, and this measurement was derived from the intensity contour: intensity difference (*IntDiff*). *IntDiff* is the intensity difference between the intensity minimum during the consonant and the intensity maximum in the following tautosyllabic vowel (Hualde et al., 2011). The more lenited the consonant, the smaller the *IntDiff* (dB) is expected to be with respect to the following vowel. Previous studies have found that *IntDiff* correlates with articulatory data (Hualde, Nadeu, & Simonet, 2010; Parrell, 2010).

Results

The *IntDiff* data are plotted in Figure 3. The figure includes three panels, one for each place of articulation. The light gray bars represent the post-pausal phrase-initial productions and the dark gray bars are the intervocalic realizations. The lower *IntDiff* values indicate a more lenited consonant. For all three places of articulation, the *IntDiff* is larger for phrase-initial voiced stops than for intervocalic voiced stops for each speaker group, demonstrating that these bilinguals are applying the spirantization rule in Spanish. The figure also shows, as expected, that the magnitude of the difference between the phrase-initial and intervocalic stops in terms of their *IntDiff* values in each position is larger for the early simultaneous and sequential bilinguals than for late L2 Spanish learners.

The *IntDiff* data were analyzed through mixed-model analyses of variance with group (L2 Spanish, simultaneous bilingual, and sequential bilingual) as between-subjects factor, point of articulation (bilabial, dental, velar) and position (phrase-initial, intervocalic) as within-subjects factors, and individual speaker as the error term. The analysis yielded a main effect of *speaker group* ($F(2,27) = 4.19$, $p < 0.05$), *consonant* ($F(2,135) = 5.48$, $p < 0.001$) and *position* ($F(1,135) = 452.13$, $p < 0.001$). The model also yielded a significant interaction between *speaker group* and *position* ($F(2,135) = 12.73$, $p < 0.001$), and between *consonant* and *position* ($F(2,135) = 7.88$, $p < 0.001$). Because the primary goal was to examine the degree of lenition in the production of the stop/spirant allophonic alternation, the main effects of speaker group were explored in a series of Bonferroni-corrected pairwise comparisons for phrase-initial and intervocalic voiced stops separately. In phrase-initial position, paired *t*-tests revealed no significant differences in terms of spirantization between the acoustic realization of sequential and simultaneous bilinguals (*diff.* = 3.8, $t(29) = 1.94$, n.s), between simultaneous bilinguals and L2 learners (*diff.* =-1.78, $t(29) = -0.97$, n.s) and between sequential bilinguals and L2 learners (*diff.* =2.02, $t(29) = 0.85$, n.s). In contrast to the results of the phrase-initial acoustic realizations, the analyses of the intervocalic voiced stops yielded significant differences between sequential and simultaneous bilinguals (*diff.* = 3.4, $t(29) = 4.57$, $p < 0.001$), between simultaneous bilinguals and L2 learners (*diff.* =6.2, $t(29) = 10.19$, $p < 0.001$) and between sequential bilinguals and L2 learners (*diff.* =9.67, $t(29) = 14.59$, $p < 0.001$).

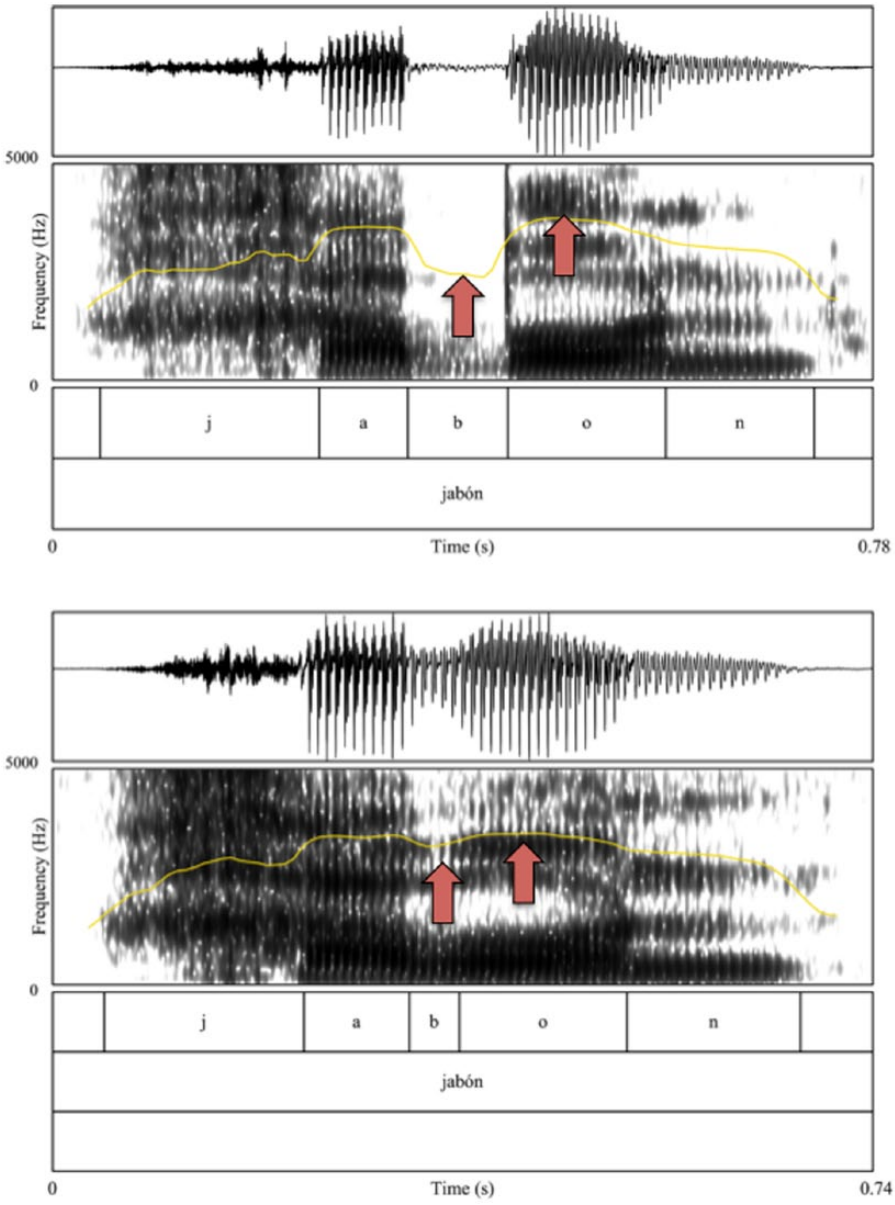

Figure 2. Waveforms and spectrograms with intensity contour of the utterance "jabón" 'soap'. The arrows indicate the points of maximum and minimum intensity of the intervocalic consonant. The intensity contour in the top waveform and spectrogram shows a more constricted realization than the waveform and spectrogram in the bottom waveform and spectrogram, which shows a more lenited acoustic realization.

Based on these results, the main effects of speaker group in intervocalic position (i.e., the context that is expected to spirantize) were investigated for each consonant separately. Figure 4 shows a plot of the *IntDiff* data in intervocalic position. The boxplot includes three panels, one for each place of articulation. For all three places of articulation, *IntDiff* is greater (showing a more

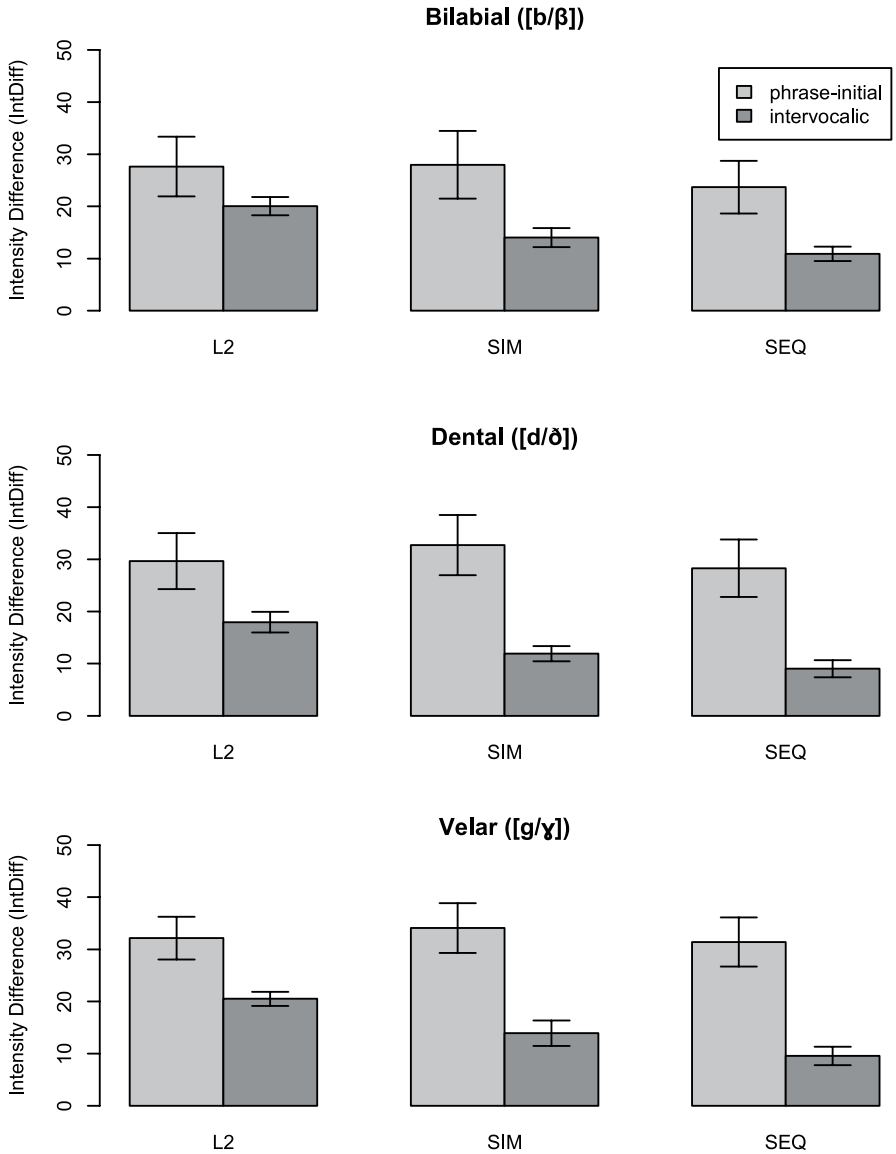

Figure 3. Intensity difference (*IntDiff*) between intensity in the consonantal minimum and intensity in the vocalic maximum of phrase-initial and intervocalic /b, d, g/ as a function of place of articulation.

constricted realization) in the production of the L2 Spanish speakers, followed by early simultaneous bilinguals, and early sequential bilinguals.

The L2 speaker group displayed significantly larger *IntDiff* values in comparison to the early simultaneous and sequential bilingual groups for each point of articulation (bilabial, dental, and velar). The production of early simultaneous and sequential bilinguals was further explored in a series of *post-hoc* analyses. Bonferroni-corrected pairwise comparisons yielded significant differences between sequential and simultaneous bilinguals in the degree of spirantization of the bilabial (*diff.* =-3.14, $t(9) = -2.51$, $p < 0.05$), dental (*diff.* =-2.9, $t(9) = -6.91$, $p < 0.01$) and velar intervocalic

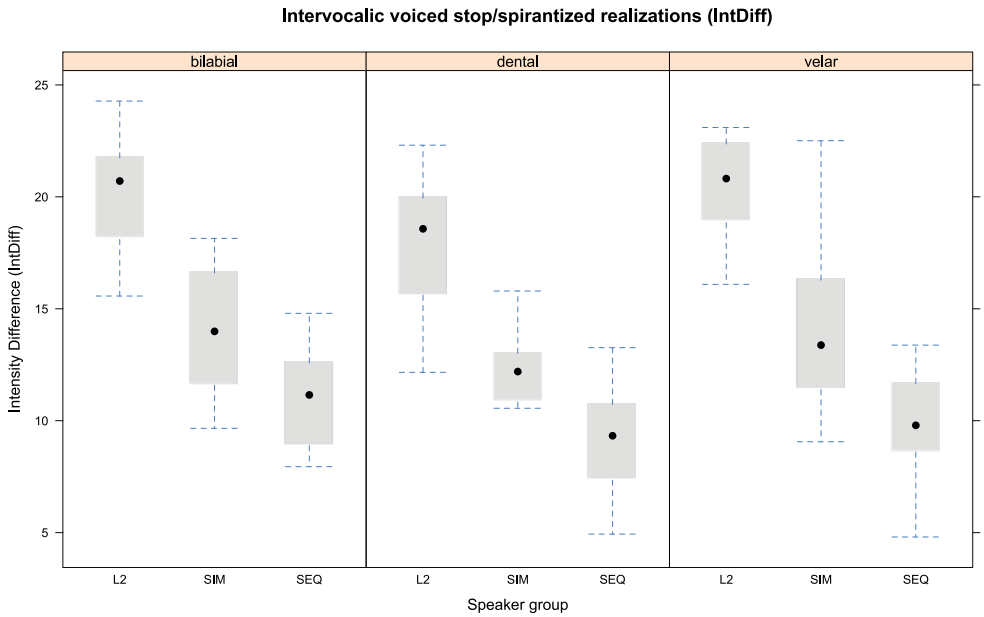

Figure 4. Boxplots of intensity difference (*IntDiff*) realized in intervocalic position as a function of place of articulation.

consonants (*diff.* =-4.37, $t(9)$ =-2.59, $p < 0.01$). These results confirm that early sequential bilinguals have smaller *IntDiff* values, therefore producing a more lenited realization of the intervocalic consonant than simultaneous bilinguals and L2 Spanish speakers. Crucially, the sequential bilingual speakers' acoustic realization of the stop/approximant alternation in Spanish is different from that of early simultaneous bilinguals, despite having comparable language dominance, Spanish language use, and self-reported Spanish accent. These results suggest that early exposure and type of early bilingualism has a profound influence on the way native sounds are articulated at the level of fine-grained phonetic detail.

Discussion and conclusions

The present study examined the acoustic realization of phrase-initial /b, d, g/ and intervocalic [β, ð, ɣ] in the production of English-dominant L1 English–L2 Spanish learners and early Spanish-English bilinguals (Spanish heritage speakers). The group of early bilinguals was further divided into simultaneous and sequential bilinguals based on the type of early bilingualism: simultaneous bilinguals had acquired both of their languages at the same time in early childhood whereas the L2 of sequential bilinguals (English) had been acquired after the structural foundations of the L1 (Spanish) were in place (Montrul, 2008). The main goal of this production experiment was to investigate if the type of early bilingualism has consequences on the acoustic realization of language-specific phonological processes, such as Spanish intervocalic spirantization, in adulthood.

The results of the reading-aloud task indicate that even though all speakers produce a more spirantized consonant in intervocalic position than in phrase-initial position, showing that they successfully apply this Spanish spirantization rule, early bilinguals (i.e., Spanish heritage speakers)

produce a more lenited intervocalic /b, d, g/ than late L2 Spanish learners. At least in terms of spirantization[2], no significant differences were found between the three speaker groups in the acoustic realization of phrase-initial voiced stops. Furthermore, sequential bilinguals are performing differently than simultaneous bilinguals with respect to the degree of lenition in intervocalic position by producing a more lenited intervocalic bilabial, dental, and velar consonant than simultaneous bilinguals. These results suggest that for early bilinguals, who have been exposed to both the minority (i.e., heritage) language and the majority language early in life, the effects of the formative first years of life seems extremely persistent, and they are carried into adulthood. The novel contribution of the current study is that it shows that the speech production of early simultaneous and sequential bilinguals is also shaped by the quantity of L1 and L2 use during these developmental stages as a result of the type of early bilingualism. In other words, differences in early exposure have consequences in the production of language-specific acoustic targets.

The results of this study are in line with the findings in Sebastián-Gallés et al. (2005), suggesting that even the slightest variation in the amount of language exposure during the earliest stages of language development can affect an individual's perceptual abilities into adulthood. The present study is the first experiment to directly investigate the effects of the type of early bilingualism on the production of a phonetic variable in the heritage language as an adult. However, it is not the first to show the critical importance of the first years of life on an individual's phonological development. Several studies have examined the speech production abilities of adults who, as children, were exposed to a language without ever acquiring it (Au, Knightly, Jun, & Oh, 2002; Casillas & Simonet, 2016; Knightly et al., 2003; Oh, Jun, Knightly, & Au, 2003). For instance, Knightly et al. (2003) showed that adults, who were not able to speak Spanish but who were passively exposed to the language as children, had more native-like Spanish pronunciation than typical late L2 learners. Interestingly, the childhood overhearing advantage seems to be domain specific: it was robust in phonology, a domain that is easy for children to acquire but difficult for adults to master, but was not detectable in morphosyntax (Au et al., 2002). Casillas and Simonet (2016) recently examined the production and perception of the /æ/-/ɑ/ vowel contrast by a group of proficient English speakers who were raised by Spanish-speaking families but who became dominant in English during childhood and, as adults, lacked communicative abilities in Spanish, together with a group of Spanish-speaking late learners of English who were dominant in Spanish. Both experimental groups differed from native controls in their production and perception of the English-specific vowel contrast. Their results also point to the conclusion that the formative first years of life appear to have a lifelong impact on phonetic behavior, and this seems to hold even when these speakers no longer have productive abilities in the language.

It is possible that the simultaneous bilinguals in the present study may have had less exposure to Spanish during childhood, adolescence, and early adulthood than their sequential bilingual peers, due to the social environment in which they grew up. This experiment does not directly test this possibility, even though it is highly unlikely given that they did not differ in their frequency and context of use, self-reported proficiency in speaking, listening, reading, and writing, or their attitudes towards each language. Participants' language dominance scores based on their responses to the BLP were comparable: sequential bilinguals (mean (M) = 29.6, standard deviation (SD) = 9.6) versus simultaneous bilinguals (M = 32.8, SD = 10.1). With this caveat in mind, these findings provide further evidence that heritage speakers are a heterogeneous group that varies greatly even if they acquire both languages early in life, and that simultaneous bilinguals and L2 learners differ from sequential bilinguals in the production of the stop/approximant allophonic alternation in Spanish. In order to account for the patterns of performance in this experiment, a model of L2 phonological acquisition must not only take into consideration the age of

acquisition, but also the type of early bilingualism, and the quantity and quality of the ambient input that is received early in life.

There are several factors that have not been included in the design of the current study and future work will benefit from exploring these variables. For instance, the effects of task type, where we might find differences in a reading task versus spontaneous production (Rao, 2015), or cognate status, where we might expect more cross-linguistic influence from English in cognates versus non-cognates (Amengual, 2012). This study was also not designed to test the effect of sociolinguistic factors with data from heritage speakers of different ages, genders, and national origins. Future studies on the phonetic abilities of Spanish heritage speakers should explore more extralinguistic variables while also taking into account the linguistic background of these heritage speakers' parents to obtain a clearer picture of the baseline variety of Spanish to which these heritage speakers have been exposed to as children. While much work remains to be done to fully understand the phonological/phonetic system of heritage speakers, the findings of this study add to our understanding of the sources of variation in early bilingual pronunciation, showing that the type of bilingual language development, at a very early age, has consequences on the acoustic realization of allophonic variants in adulthood.

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Notes

1. These weakened allophones have been classified as fricatives (Navarro Tomás, 1967; Quilis, 1993) or approximants (Hualde, 2005; Martínez Celdrán, 1985), with the latter scholars arguing that they should not be considered fricatives because they do not exhibit the aperodic energy, lengthy duration, and relatively low intensity that characterize fricatives. In this study, I will refer to these segments as approximants because the distance between articulators is wide enough to prevent turbulent airflow. In this case, consonantal tension has an inverse relationship with intensity: the energy increases in a gradient fashion as the distance between articulators increases (Hualde, 2005; Martínez Celdrán, 1985).
2. It is acknowledged that *IntDiff* is not an acoustic correlate typically used to examine phrase-initial voiced stops. This measurement is included to serve as a point of comparison with the acoustic realization of these segments in intervocalic position. In order to more closely investigate phrase-initial stops, future studies should include other acoustic correlates, such as voice onset time or onset of fundamental frequency.

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